General Dessert #11: Speech Processors, S Meters, and Operating Near Band Edges

Speech-processor purpose and pitfalls, S-meter calibration (6 dB / S-unit), the 4x rule from S8 to S9, and the 3 kHz LSB/USB carrier-offset rules for staying inside the phone segment.

Group G4D · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Speech Processor

Question	Answer	Q
Purpose of a speech processor	**Increase the apparent loudness of transmitted voice signals.**	G4D01
Effect on an SSB phone signal	**Increases average power** (peak is already at PEP; compression raises the mean).	G4D02
Effect of incorrect adjustment	**All of the above**: distorted speech, excess IMD, excessive background noise.	G4D03

S Meter and Signal Strength

Question	Answer	Q
What does an S meter measure?	**Received signal strength.**	G4D04
How much power change per S-unit?	**6 dB** (one S-unit, by convention).	G4D06
How does “20 dB over S9” compare to “S9”?	**100 times more powerful** (20 dB = 10^2).	G4D05
Power increase needed to go S8 to S9 on a distant receiver	**Approximately 4 times** (6 dB).	G4D07

Operating Near Band Edges (3 kHz SSB)

Sideband geometry: LSB places voice **below** the displayed carrier frequency, USB places voice **above**. A 3 kHz signal occupies 3 kHz on the active side of the carrier.

Setup	Occupied range / margin	Q
3 kHz LSB, carrier at 7.178 MHz	**7.175 to 7.178 MHz** (3 kHz below carrier).	G4D08
3 kHz USB, carrier at 14.347 MHz	**14.347 to 14.350 MHz** (3 kHz above carrier).	G4D09
LSB near lower phone-band edge	Carrier **at least 3 kHz above** the lower edge of the segment.	G4D10
USB near upper phone-band edge	Carrier **at least 3 kHz below** the upper edge of the band.	G4D11

Dessert 11 Question Bank (G4D)

G4D01. What is the purpose of a speech processor in a transceiver?
A. Increase the apparent loudness of transmitted voice signals
B. Increase transmitter bass response for more natural-sounding SSB signals
C. Prevent distortion of voice signals
D. Decrease high-frequency voice output to prevent out-of-band operation

G4D02. How does a speech processor affect a single sideband phone signal?
A. It increases peak power
B. It increases average power
C. It reduces harmonic distortion
D. It reduces intermodulation distortion

G4D03. What is the effect of an incorrectly adjusted speech processor?
A. Distorted speech
B. Excess intermodulation products
C. Excessive background noise
D. All these choices are correct

G4D04. What does an S meter measure?
A. Carrier suppression
B. Impedance
C. Received signal strength
D. Transmitter power output

G4D05. How does a signal that reads 20 dB over S9 compare to one that reads S9 on a receiver, assuming a properly calibrated S meter?
A. It is 10 times less powerful
B. It is 20 times less powerful
C. It is 20 times more powerful
D. It is 100 times more powerful

G4D06. How much change in signal strength is typically represented by one S unit?
A. 6 dB
B. 12 dB
C. 15 dB
D. 18 dB

G4D07. How much must the power output of a transmitter be raised to change the S meter reading on a distant receiver from S8 to S9?
A. Approximately 1.5 times
B. Approximately 2 times
C. Approximately 4 times
D. Approximately 8 times

G4D08. What frequency range is occupied by a 3 kHz LSB signal when the displayed carrier frequency is set to 7.178 MHz?
A. 7.178 MHz to 7.181 MHz
B. 7.178 MHz to 7.184 MHz
C. 7.175 MHz to 7.178 MHz
D. 7.1765 MHz to 7.1795 MHz

G4D09. What frequency range is occupied by a 3 kHz USB signal with the displayed carrier frequency set to 14.347 MHz?
A. 14.347 MHz to 14.647 MHz
B. 14.347 MHz to 14.350 MHz
C. 14.344 MHz to 14.347 MHz
D. 14.3455 MHz to 14.3485 MHz

G4D10. How close to the lower edge of a band's phone segment should your displayed carrier frequency be when using 3 kHz wide LSB?
A. At least 3 kHz above the edge of the segment
B. At least 3 kHz below the edge of the segment
C. At least 1 kHz below the edge of the segment
D. At least 1 kHz above the edge of the segment

G4D11. How close to the upper edge of a band's phone segment should your displayed carrier frequency be when using 3 kHz wide USB?
A. At least 3 kHz above the edge of the band
B. At least 3 kHz below the edge of the band
C. At least 1 kHz above the edge of the segment
D. At least 1 kHz below the edge of the segment

Answer Key

G4D01: A	G4D02: B	G4D03: D	G4D04: C
G4D05: D	G4D06: A	G4D07: C	G4D08: C
G4D09: B	G4D10: A	G4D11: B	